

Project Blueprint — Curation Integrity Audit for NeMo Curator

Prepared for the TCS expert consultation

September 2026

Contents

0. Read This First	2
1. Purpose	3
1.1 The problem in plain language	3
1.2 Why that question suddenly matters	3
1.3 What we are actually building	3
2. Target Field of Work	4
2.1 Where this sits in the AI lifecycle	4
2.2 Domains this applies to	4
2.3 Industry verticals where it bites hardest	4
3. The Evidence Base	4
3.1 Finding 1 — Quality scorers prefer machine-written text	4
3.2 Finding 2 — Quality filters penalise non-Western content	5
3.3 Finding 3 — Well-written poison survives curation	5
3.4 The unifying insight	5
3.5 What is genuinely unknown — and why that’s your project	6
4. What Already Exists (Know This Or Get Caught)	6
4.1 NeMo Curator itself	6
4.2 CuratorKIT — the adjacent work you must acknowledge	6
4.3 Related research areas	7
4.4 The honest positioning	7
5. How This Benefits TCS	7
5.1 The direct business case	7
5.2 The risk-avoidance case	7
5.3 The strategic case	8

5.4	The India-specific case	8
6.	Why Us — The Honest Version	8
6.1	Don't claim superiority. Claim availability.	8
6.2	The killer line: don't trust us, check the output	8
6.3	What you actually bring	8
6.4	What you should openly admit	9
7.	Technical Workflow	9
7.1	The full architecture.	9
7.2	Why probe sets are the right method	9
7.3	Build order (do not reorder this)	10
7.4	The ordering detail that signals competence	10
8.	Data and Compute	10
8.1	Data you need, and where to get it	10
8.2	Compute requirements — the honest assessment	11
8.3	Environment notes	11
8.4	What to ask TCS for	11
9.	The Demonstration	12
9.1	The one chart that carries the whole project	12
9.2	What NOT to do tomorrow	12
10.	Risks and Honest Gaps	12
10.1	The findings might not transfer	13
10.2	Confounding is the main methodological threat	13
10.3	"Bias" needs a careful definition	13
10.4	Scale limits	13
10.5	One unverified number	13
10.6	Overlap risk	13
11.	Tomorrow's Meeting Playbook	13
11.1	Structure	13
11.2	Questions to ask them	14
11.3	Handling the likely pushbacks	14
11.4	Tone	14

12. Background You Should Know	15
12.1 Glossary	15
12.2 The four things to memorise	15
12.3 If you only have two hours tonight	16
13. Final Note	16

0. Read This First

This document is written so you can walk into tomorrow’s meeting having read nothing else.

What this project is, in one sentence:

We build an India-tuned NeMo Curator pipeline for enterprise data, plus a measurement harness that proves what the pipeline keeps and what it silently throws away.

Two framings of the same work. Which one you lead with depends on the room.

Framing	Use when	Sounds like
A — Application-first (safe)	They want NeMo Curator <i>used</i> in a system	Indian enterprise data and validating the tuning”
B — Insight-first	They engage with the (ambitious) research finding “Curation filters are unaudited; we’re building the audit”	

Default to A. Let B emerge on slide 4. If you lead with B it sounds like criticism of a tool they may already use. If you lead with A it sounds like diligence, and B arrives as your justification rather than your attack.

1. Purpose

1.1 The problem in plain language

Before you train or fine-tune an AI model, you clean the data. NeMo Curator is NVIDIA’s tool for doing that at scale — it removes duplicates, strips out personal information, and filters out low-quality text.

The filtering step uses a **quality classifier**: a model that scores each document and drops the ones below a threshold.

Nobody checks what that classifier is biased toward.

It is the single most consequential component in the pipeline — it decides what your model will and won’t ever see — and it is the least examined. Teams tune dedup thresholds, benchmark throughput, and measure downstream accuracy. Almost nobody asks: *what kind of text does this filter systematically prefer, and what kind does it systematically destroy?*

1.2 Why that question suddenly matters

Two things changed.

The web filled with AI-generated text. If the filter prefers machine-written text, then scaling up curation doesn't remove synthetic content — it concentrates it. The tool meant to protect training data quality becomes the mechanism that degrades it.

LLM development went global. Filters were built and validated on Western English web text. When you point them at Indian English, Indic-language content, or enterprise documents written by non-native speakers, “different style” can read as “low quality.”

1.3 What we are actually building

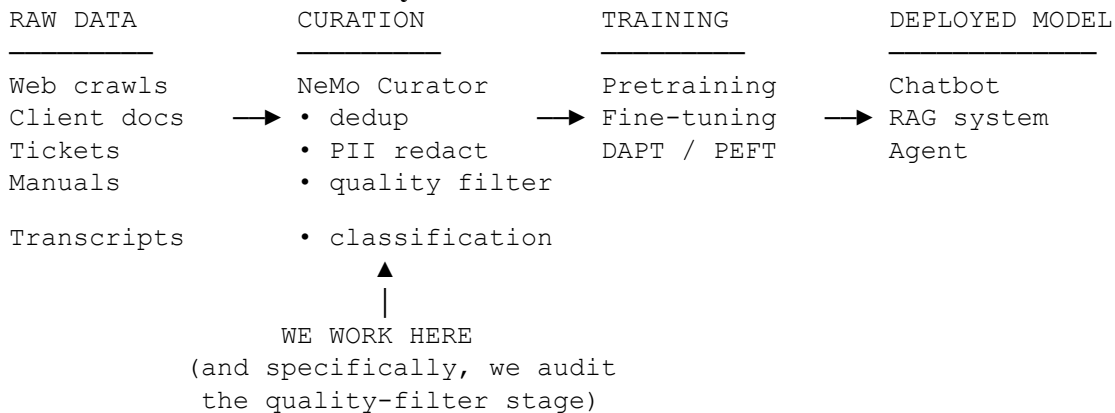
Three deliverables, in dependency order:

1. **A working NeMo Curator pipeline** configured for Indian enterprise text — the thing that runs.
2. **A probe-based audit harness** — feed in labelled document groups, measure the survival rate of each group through each stage, output a report card.
3. **A tuned configuration** — thresholds and classifier choices adjusted based on what the audit reveals, reusable across engagements.

Deliverable 1 is the application. Deliverable 2 is the novelty. Deliverable 3 is the thing TCS could actually put on a shelf and reuse.

2. Target Field of Work

2.1 Where this sits in the AI lifecycle



Everyone competes at the training and deployment stages. The curation stage is upstream of both, less crowded, and determines the ceiling on everything downstream. A model cannot learn what curation deleted.

2.2 Domains this applies to

- **Domain-adaptive pretraining (DAPT)** — adapting a general model to banking, healthcare, telecom
- **Supervised fine-tuning (SFT)** — instruction data curation
- **RAG corpus preparation** — retrieval knowledge bases, which nobody curates properly
- **Multilingual and Indic model development** — where the bias problem is sharpest

2.3 Industry verticals where it bites hardest

Regulated, document-heavy, non-Western-English industries — which is a precise description of TCS’s client base. Banking and financial services, healthcare, government and public sector, telecom.

3. The Evidence Base

This section is the spine of your pitch. Know these three findings.

3.1 Finding 1 — Quality scorers prefer machine-written text

A study published in *ACM Transactions on Information Systems* benchmarked text quality scoring models. The result: LLMs and transformer pretrained language models score GPT-generated text **10–20% higher on average** than human-authored documents. Traditional feature-based ML models showed the opposite — they scored human text considerably higher.

Why this matters: NeMo Curator’s quality classifier is `nvidia/quality-classifier-deberta` — a transformer PLM, exactly the category that shows the bias.

The implication chain:

Web is increasingly AI-generated



Quality filter prefers AI-generated text (+10-20%)



Curation concentrates synthetic content rather than removing it



Next model trains on more synthetic data than the raw web contained ▼

Model collapse, accelerated by the tool meant to prevent it

Status: peer-reviewed, solid. Cite it confidently.

3.2 Finding 2 — Quality filters penalise non-Western content

Filters trained on English web data encode “good writing” as Western, academic, and formal. Reported effect: they downgrade content from non-Western sources — blogs from India, forums from Brazil, technical documentation in Arabic — by **22–27%**.

Why this matters to TCS specifically: they are an Indian company, building AI for Indian clients, on Indian data. If the industry-standard curation stack discards roughly a quarter of Indian-authored content as low quality, then every India-focused model built on a default config starts from a corpus that has been culturally strip-mined — before a single training step runs.

⚠ Status: VERIFY BEFORE YOU PUT THIS ON A SLIDE. This figure came to us via a secondary source summarising research, not the primary paper. Find the original study or drop the specific number and state the effect qualitatively. Putting an unverified statistic in front of experts is the fastest way to lose the room.

3.3 Finding 3 — Well-written poison survives curation

Research on persistent pre-training poisoning argues that attacks such as context extraction and belief manipulation are likely to bypass most filters, because they are written in fluent English and don’t exhibit the artifacts filters actually

target, like HTML markup. A separate study measured a **0.13% infiltration rate** — poisoned content that curation misses entirely.

Why: filters look at *how text is written*, not *who wrote it or why*. A competent attacker writing clean prose is indistinguishable from a competent author.

Status: well-supported in the literature. This one is established enough that TCS experts may already know it.

3.4 The unifying insight

All three failures have one root cause:

The quality classifier does not measure quality. It measures style.

Fluency, formality, and structure are what it can see. Truth, provenance, originality, and cultural context are what it cannot.

Everything follows:

- AI text is stylistically smooth → **boosted**
- Indian English differs stylistically from Western academic English → **cut**
- Well-crafted poison is stylistically clean → **admitted**

If you remember one line from this document, make it that one.

3.5 What is genuinely unknown — and why that’s your project

Every finding above concerns transformer quality scorers **as a category**. To the best of our research, **nobody has published an audit of NVIDIA’s specific quality-classifier deberta for either bias**.

That gap is the project. The honest framing:

“The literature establishes that transformer quality scorers exhibit these biases. NeMo Curator ships a transformer quality scorer. Nobody has tested whether it inherits them. We want to measure it.”

You are not claiming to have found a flaw. You are pointing out that **nobody has looked** — which is true, defensible, and the kind of question experts enjoy being handed.

4. What Already Exists (Know This Or Get Caught)

Walking in without knowing the adjacent work is how you lose credibility. Here is the landscape.

4.1 NeMo Curator itself

NVIDIA’s GPU-accelerated curation toolkit. Confirmed current capabilities:

- Exact, fuzzy, and semantic deduplication
- Heuristic and classifier-based quality filtering
- PII detection and redaction
- Language identification, domain and safety classification
- Downstream-task decontamination
- Synthetic data generation (SDG)
- Built on RAPIDS (cuDF, cuML, cuGraph) with Ray, scaling laptop → multi-node GPU

Proven at scale: it powers the data pipelines behind NVIDIA’s Nemotron models, and the Nemotron-4 pretraining dataset was curated with its text pipeline across 8+ trillion tokens of multilingual web data.

Critical version note: release 26.02 moved to a Ray-based pipeline architecture across text, image, video, and audio. Version 1.x was a **major rewrite** — the old `DocumentDataset` and `nemo_curator.modules.*` API from 0.x is gone. You now compose `ProcessingStages` into a `Pipeline` and run it with an executor.

Most tutorials online are still 0.x. If you paste 0.x imports into a demo, TCS experts will notice instantly. Follow the current quickstart, not blog posts.

4.2 CuratorKIT — the adjacent work you must acknowledge

An open-source library published in 2026 that builds auditable post-training curation pipelines. Its stated motivation: existing tools treat ingestion, deduplication, synthetic generation, and quality filtering as separate stages, and that fragmentation makes it hard to audit pipeline decisions or understand why individual samples are rejected.

This matters because it overlaps our earlier idea. An earlier version of this pitch proposed building a per-document audit trail and provenance manifest. CuratorKIT substantially covers that. We changed direction because of it.

How to handle it if raised: *“CuratorKIT audits which samples were rejected. We’re asking a different question — not ‘which documents were dropped’ but ‘which populations were disproportionately dropped.’ Their unit of analysis is the document. Ours is the demographic group.”*

That distinction is real and it holds up.

4.3 Related research areas

- **Datapoisoning/backdoorattacks** — mature field. OWASP lists training data poisoning as LLM03; NIST AI 100-2 classifies it as a key adversarial ML threat.
- **Model collapse** — well-established that recursive training on synthetic output degrades models.
- **Machine-generated text detection** — active field; some work explicitly frames detection as a collapse-prevention mechanism.
- **Classifier-based quality filtering skepticism** — at least one 2025 paper questions whether classifier-based quality filtering delivers what practitioners assume.

Read that last one before Friday if you have time. It is the closest existing work to your thesis, and you should know whether it already answers your question.

4.4 The honest positioning

You are not inventing a field. You are combining three known results and pointing them at a specific, widely-deployed, unaudited tool — with an India-specific angle nobody has commercial incentive to investigate.

That is a legitimate contribution. Don’t oversell it as more.

5. How This Benefits TCS

5.1 The direct business case

TCS runs many client engagements simultaneously, each needing data prepared for fine-tuning or RAG. Right now, each engagement tunes curation by hand, with no way to know what the config is discarding.

This project produces a **reusable artifact**: a validated curation configuration plus a test harness that proves it behaves correctly on Indian enterprise data. That converts a per-project judgment call into a standardised, defensible process.

Services companies make money on repeatability. A curation config that has been measured and can be justified to a client is worth more than one that merely runs.

5.2 The risk-avoidance case

If the bias effects hold, then any TCS model built on a default curation config has been trained on a corpus that systematically under-represents Indian-authored content. Consequences:

- Models that perform worse on the Indian English their end users actually write
- A compliance and reputational exposure if a client ever asks what was filtered and why
- Silent quality degradation nobody can diagnose, because the deleted data left no trace Finding this before deployment is far cheaper than finding it after.

5.3 The strategic case

If synthetic content is being amplified rather than filtered by standard pipelines, every organisation training on web-derived data has a problem arriving on a schedule. The organisation that can **measure** its exposure has a genuine advantage over those who cannot.

5.4 The India-specific case

This is the argument that lands hardest in this particular room. India is building sovereign AI capability. The curation layer of that stack is currently imported wholesale from tooling validated on Western English. Nobody has checked whether it fits.

TCS is unusually well-positioned to care: Indian company, Indian data, Indian clients, Indian regulatory environment. A foreign vendor has no commercial incentive to discover that their filter downranks Indian content. TCS does.

6. Why Us — The Honest Version

You will be asked some version of “*why should we have students do this rather than our own people?*” Do not bluff. Here is the answer that actually works.

6.1 Don’t claim superiority. Claim availability.

TCS experts are better at this than you are. Say so if it comes up — it costs nothing and buys credibility.

The real argument is different: **their experts are on billable client work**. Nobody gets budget to spend three weeks systematically measuring whether a filter is biased, because it isn’t attached to a deliverable. It is unglamorous, methodical, negative-result-tolerant work.

That is exactly the work a student team is well-suited to. Not because you’re smarter — because you’re available for the kind of task that never survives a prioritisation meeting.

6.2 The killer line: don’t trust us, check the output

“You shouldn’t trust us. The output is a reproducible measurement — same probe set, same config, same result. Run it yourself and see if you get our numbers. That’s the point of building it as a harness rather than a report.”

This is a genuinely strong position. You are not asking them to believe a claim. You are handing them an instrument.

6.3 What you actually bring

- **No incentive to find a comfortable answer.** A vendor auditing their own filter has an obvious conflict. You don't.
- **Willingness to publish a null result.** If the classifier turns out unbiased, that's a finding and you'll report it. Consultants rarely deliver "we checked, nothing's wrong."
- **Relevant prior work.** Your SIH project (ChainTrace) was built on exactly this discipline — measuring structure, documenting unvalidated thresholds honestly, building explainable rather than black-box outputs. That's the same muscle.

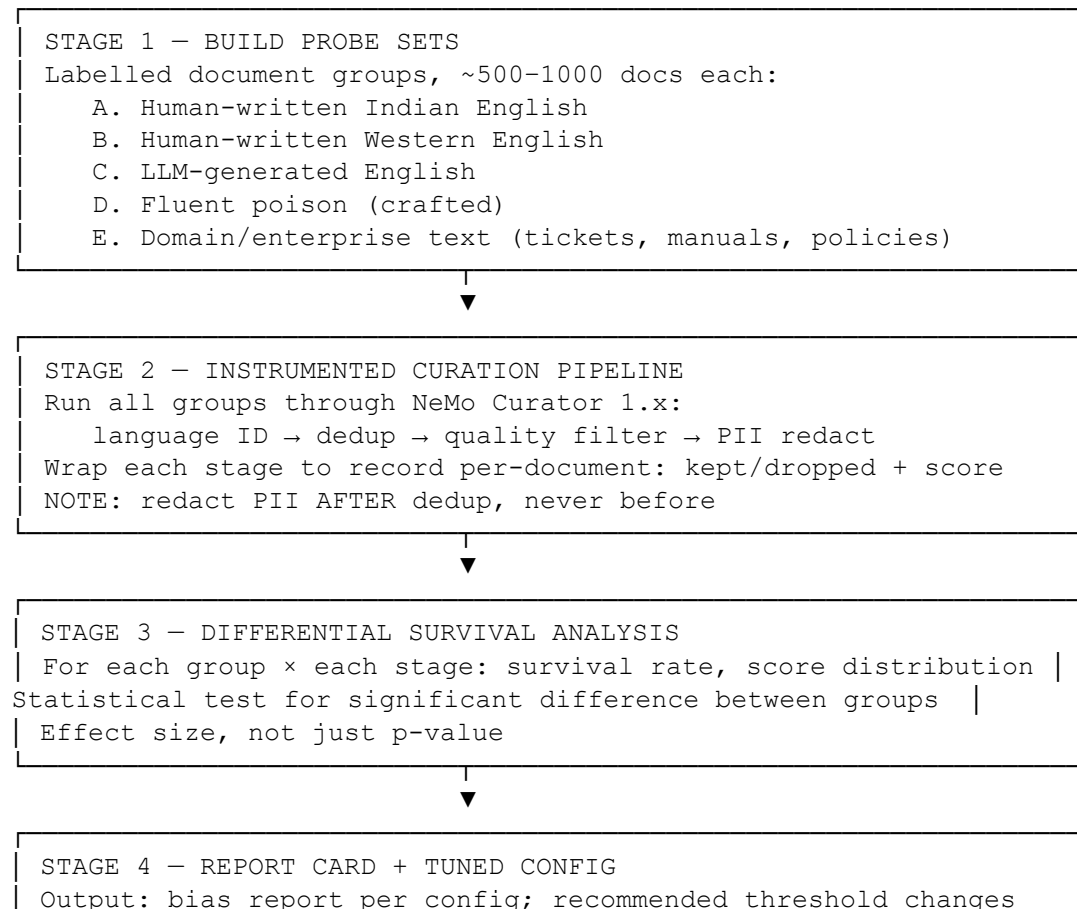
6.4 What you should openly admit

- You have not built this yet
- You do not have GPU infrastructure
- You need guidance on which enterprise data characteristics matter
- Some of your evidence still needs primary-source verification

Stating limitations up front is disarming and makes everything else you say more believable. Experts trust people who volunteer their gaps.

7. Technical Workflow

7.1 The full architecture



Re-run to verify the tuned config closes the gap

7.2 Why probe sets are the right method

You cannot audit a classifier by reading its code. You audit it the way you'd audit any black box: send in known inputs, measure what comes out.

This is standard practice in fairness testing and in security testing. You're applying an established method to a component nobody has applied it to.

The key design requirement: **groups must differ only in the attribute you're testing**. If your Indian English samples are also on different topics or from lower-quality sources than your Western samples, you've measured the wrong thing. Matching topic, length, and register across groups is the hardest and most important part of the build.

7.3 Build order (do not reorder this)

1. Get NeMo Curator installed and one trivial pipeline running end to end. **Do this first**. Installation is the most likely thing to consume a day unexpectedly.
2. Assemble the smallest useful probe set — 100 docs per group is enough to see an effect.
3. Instrument one stage only: the quality classifier. Ignore everything else initially.
4. Produce the one chart (see §9.1).
5. Only then expand to more stages, more groups, larger samples.

Resist building the full architecture first. Get one measurement working end to end, then widen.

7.4 The ordering detail that signals competence

Redact PII after deduplication, never before. Exact and fuzzy dedup depend on document content being consistent; redacting first can break hash-based dedup because PII strings get replaced inconsistently.

Mention this naturally in the meeting. It's a small thing that demonstrates you read the actual documentation rather than a summary.

8. Data and Compute

8.1 Data you need, and where to get it

Probe group	Candidate sources	Notes
Indian English (human)	AI4Bharat corpora, Indian news archives, Indian academic/technical writing, government publications	[VERIFY] licensing before use
Western English (human)	Wikipedia, C4, FineWeb samples	Widely available, permissive
LLM-generated	Generate yourself with an open model; or existing human-vs-machine comparison datasets	Generating your own gives you control over topic matching

Fluent poison	Craft manually — 20–50 documents is plenty	Keep them benign in content; you’re testing filter behaviour, not building real attacks
Enterprise text	Ask TCS — see §8.4	The one thing you cannot source yourself

Critical methodological point: for the Indian vs Western comparison, control the topic. Write or select documents on the *same subjects* in both varieties. Otherwise you’re measuring topic bias, not dialect bias, and your result is worthless.

8.2 Compute requirements — the honest assessment

For the initial demo: a laptop is enough.

The quality classifier is a DeBERTa-class transformer. Scoring a few hundred documents on CPU takes minutes, not hours. NeMo Curator ships CPU extras (`text_cpu`) precisely so you can develop without GPUs.

What genuinely needs GPU:

- Semantic deduplication at scale (requires embedding every document)
 - Fuzzy dedup on large corpora
 - Any run above roughly 10^5 documents
 - Anything approaching realistic enterprise corpus size
- Recommended honest position for the meeting:**

“The measurement itself runs on CPU at probe-set scale — we can produce the first result on a laptop. We’d need GPU access to validate at realistic corpus scale, and that’s one of the things we’d want to discuss.”

That is a reasonable, non-greedy ask, and it shows you understand the actual cost structure.

8.3 Environment notes

- Install: `pip install "nemo-curator[text_cpu]"` for the CPU path
- Video and audio pipelines depend on system codec libraries; the published container ships them preconfigured — irrelevant for you, but know it exists
- The NeMo Framework Container is the path of least resistance if local install fights you
- [VERIFY] exact `1.x` stage names and imports against the current quickstart before writing any code

8.4 What to ask TCS for

Ask for these in priority order. Asking for concrete, modest things reads far better than asking for “support.”

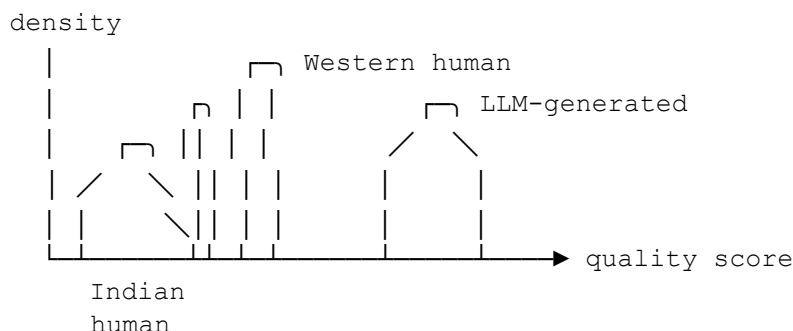
1. **Guidance on what enterprise text actually looks like** — even a description, not data. What proportion is tickets vs manuals vs transcripts? How non-native is the English? This shapes your probe set and costs them nothing.
2. **A sanitised or synthetic sample of representative enterprise documents.** Frame it as optional and note you can proceed with public proxies if data sharing is complicated. Do not put them in an awkward position.
3. **GPU access** for the scale-up phase, if the initial result is interesting.
4. **A reviewer** — someone who’d spend 30 minutes checking your methodology before you run the full study. This is the cheapest and most valuable thing you can ask for, and it flatters them appropriately.

Lead with request 1 and 4. They cost nothing and are easy to say yes to. A meeting that ends with a small commitment is far better than one that ends with a large refusal.

9. The Demonstration

9.1 The one chart that carries the whole project

Take three groups of documents — human Indian English, human Western English, LLM-generated — matched on topic and length. Run all through NeMo Curator’s quality classifier. Plot the three score distributions on one axis. Quality score distribution by group



If it looks like this, you have proven the finding in one image.

Three possible outcomes, all publishable:

Outcome	What it means	What you say
Indian lower, AI higher	Both biases confirmed in NVIDIA’s classifier	Strong finding — this becomes the project
One holds, one doesn’t	Partial confirmation	Still novel; narrows the question usefully
Neither holds	NVIDIA’s classifier avoids the documented category bias	Also a real finding. Report it honestly.

You genuinely cannot lose this experiment. Say that in the meeting — it demonstrates you’ve designed something falsifiable, which is what separates research from advocacy.

9.2 What NOT to do tomorrow

You have no demo. **Do not attempt to build one tonight.** A half-working prototype presented to experts is strictly worse than an honest plan.

Nobody expects code at a problem-statement scoping meeting. What they expect is that you understand the domain and have thought about methodology. Spend the evening reading, not debugging an install.

10. Risks and Honest Gaps

State these yourself before someone finds them. Volunteering limitations is the single most credibility-positive thing you can do in a room of experts.

10.1 The findings might not transfer

The research concerns transformer quality scorers generally. NVIDIA may have specifically trained against these biases. **This is unknown, and it’s the entire reason to run the study.**

10.2 Confounding is the main methodological threat

If Indian and Western probe documents differ in topic, source quality, or length, you measure the wrong thing. Topic-matching is essential and genuinely difficult. This is where a TCS reviewer would add the most value.

10.3 “Bias” needs a careful definition

Lower quality scores are not automatically unfair. If a group’s documents genuinely are lower quality by some defensible standard, downranking them is correct behaviour. Your claim must be narrower: **stylistic difference is being scored as quality difference**. Keep the claim tight.

10.4 Scale limits

Laptop-scale probe sets demonstrate an effect. They do not establish what happens at trilliontoken scale, where distributions and thresholds behave differently. Be explicit about that boundary.

10.5 One unverified number

The 22–27% non-Western downranking figure requires primary-source confirmation. Either verify it before Friday or state the effect qualitatively without the number.

10.6 Overlap risk

CuratorKIT covers audit-of-rejected-samples. Your differentiator — population-level rather than document-level analysis — is real but you must be able to articulate it in one sentence under pressure.

11. Tomorrow’s Meeting Playbook

11.1 Structure

Minutes	Content
0–2	What NeMo Curator is and where it sits in the LLM lifecycle
2–4	The pipeline stages, in order
4–7	The gap: filters are unaudited; the three findings
7–9	The proposal, application-first framing
9–11	Workflow, data, compute, what we’d need
11–15	Open questions for them

Reserve real time for questions to them. This is a consultation, not a defence.

11.2 Questions to ask them

These are as important as your pitch. Good questions signal more competence than good answers.

1. Does NeMo Curator 26.02 already expose per-stage retention statistics, or would we be building that instrumentation ourselves?
2. In your client work, has anyone observed quality filters behaving badly on Indian English or non-native English enterprise text?
3. What does enterprise text actually look like in your engagements — proportions of tickets, manuals, transcripts, policy documents?

4. Is the quality classifier’s training data documented anywhere? That would tell us immediately whether these biases are plausible.
5. If we found a significant effect, what would you need to see before trusting the result?

Question 5 is the strongest one. It invites them to define your success criteria, which makes them invested.

11.3 Handling the likely pushbacks

“We wanted NeMo Curator used in a system, not critiqued.” > “That’s what this is — we run the full pipeline; we just also measure what each stage drops. The deliverable is a tuned config for Indian enterprise data plus the evidence that the tuning was necessary.”

“Our experts could do this.” > “Absolutely, and better. But this is unglamorous measurement work that doesn’t attach to a billable deliverable, so it tends not to get prioritised. That’s the gap we can fill. And the output is a reproducible harness — you don’t have to trust our numbers, you can re-run it.”

“This already exists.” > “Which part? CuratorKIT does document-level audit of rejections. We’re asking a population-level question — not which documents got dropped, but which groups got disproportionately dropped. If something covers that, we’d genuinely like the reference.”

“How do you know the classifier is biased?” > “We don’t. That’s the point — the literature documents this for transformer quality scorers as a class, and nobody has published a test of NVIDIA’s specific one. We want to measure it. A null result is a real result.”

“What if you find nothing?” > “Then we’ve established that NVIDIA’s classifier avoids a documented category-level failure, which is useful to know before deploying it on Indian data. We designed it so both outcomes are informative.”

11.4 Tone

Do not oversell. You are students proposing a measurement study to people who build this for a living. Confidence comes from knowing your material and being straight about what you don’t know — not from claiming discovery.

The strongest posture: *“here’s a question nobody has answered, here’s how we’d answer it, and here’s what we’d need help with.”*

12. Background You Should Know

12.1 Glossary

Curation — cleaning and filtering data before training. Removes duplicates, junk, PII.

Quality classifier — a model that scores documents so low scorers can be dropped. NeMo Curator ships `nvidia/quality-classifier-deberta`.

DeBERTa — a transformer architecture used for classification tasks. The category of model shown to prefer machine-generated text.

Deduplication — removing repeated content. *Exact* = identical. *Fuzzy* = near-identical (hashing based). *Semantic* = similar meaning (embedding based).

PII redaction — detecting and removing names, emails, phone numbers. Uses NER models.

NER — Named Entity Recognition. Identifies entities like people and locations in text.

PII / DPDP — India’s Digital Personal Data Protection Act governs personal data handling; relevant to any client-data curation.

DAPT — Domain-Adaptive Pretraining. Continuing pretraining on domain text to specialise a general model.

SFT / PEFT — Supervised Fine-Tuning / Parameter-Efficient Fine-Tuning. Post-training adaptation methods.

RAG — Retrieval-Augmented Generation. Model retrieves documents at query time rather than relying on training memory.

Model collapse — degradation from recursively training models on the output of previous models.

Data poisoning — injecting malicious content into training data to manipulate model behaviour. OWASP LLM03.

Backdoor attack — poison that makes a model behave normally except when a specific trigger appears.

RAPIDS — NVIDIA's GPU-accelerated data science libraries (cuDF, cuML, cuGraph). What makes NeMo Curator fast.

Ray — distributed execution framework. NeMo Curator 26.02's pipeline backbone.

Probe set — labelled inputs sent through a black-box system to characterise its behaviour.

Differential survival rate — the core metric here: what fraction of each group survives each pipeline stage.

Confounding — when an uncontrolled variable explains your result instead of the one you're testing. The main threat to this study.

12.2 The four things to memorise

1. **"The quality classifier measures style, not quality."** — your thesis in one line.
2. **Transformer quality scorers rate GPT text 10–20% higher than human text.** — ACM TOIS, peer-reviewed.
3. **Redact PII after dedup, not before.** — the competence signal.
4. **NeMo Curator 1.x is a rewrite; ProcessingStages composed into a Pipeline, not the old 0.x module API.** — keeps you from sounding out of date.

12.3 If you only have two hours tonight

- 45 min: NVIDIA's NeMo Curator key-features page and text curation pipeline page. Be able to name the stages in order.
- 30 min: find the primary source for the non-Western downranking figure, or decide to drop the number.
- 30 min: build the six slides.
- 15 min: say the pitch out loud twice. If you can't state the PS in two sentences without notes, it isn't ready.

13. Final Note

The strongest thing about this project is that it is **falsifiable**. You have designed a study where finding nothing is as publishable as finding something. That is rare in student projects, and experts will recognise it immediately.

Do not undermine it by claiming you already know the answer. You don't. Nobody does. That's why it's worth doing.